B3 Solutions: Geometry, proof, Pythagoras, similarity and trigonometry

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Rounding: lengths to 2 decimals, angles to 1 decimal of a degree, computed from unrounded intermediate values. All numerical answers were checked with a computer algebra system (SymPy).

Practice

1. (a) At P: the angle below ℓ, left of t, is 68° (vertically opposite). The angles above-left and below-right are $180° - 68° = 112°$ (straight angle).

(b) At Q: above m, right of t: 68° (F-angle with the 68° at P, $\ell \parallel m$). Below m, left of t: 68° (vertically opposite to the angle just found). Above-left and below-right: 112°.

(c) The co-interior pair is below ℓ, right of t (112°) and above m, right of t (68°): $112° + 68° = 180°$. (The pair on the left, 68° and 112°, works too.)

2. (a) $x + 2x + 3x = 180°$, so $6x = 180°$ and $x = 30°$. The angles are 30°, 60°, 90°: a right triangle (in fact a half-equilateral triangle).

(b) Way 1: the third angle is $180° - 47° - 68° = 65°$, so the exterior angle is $180° - 65° = 115°$. Way 2: exterior angle theorem, $47° + 68° = 115°$.

(c) By the exterior angle theorem $\angle A + \angle B = 130°$, and $\angle A = \angle B + 25°$. So $2\angle B + 25° = 130°$, $\angle B = 52.5°$ and $\angle A = 77.5°$.

3. (a) $(9 - 2) \cdot 180° = 1260°$.

(b) $\dfrac{10 \cdot 180°}{12} = 150°$.

(c) Each exterior angle is $180° - 156° = 24°$, and the exterior angles add up to 360°, so $n = 360/24 = 15$. (Check: $\frac{13 \cdot 180°}{15} = 156°$.)

(d) Label the convex pentagon A, B, C, D, E in order. The star consists of the segments AC, CE, EB, BD, DA, and the star's point at each vertex is the angle between the two star segments there: at A between AC and AD, at C between CA and CE, and so on. Write $\angle A, \ldots, \angle E$ for these five angles.

Let X be the intersection of AC and EB, and Y the intersection of AD and EB. On segment EB the order is E, Y, X, B.

- Triangle XCE: its angle at C is $\angle C$ (between CA and CE) and its angle at E is $\angle E$ (between EB and EC). A lies on the extension of CX beyond X, so $\angle AXE$ is an exterior angle: $\angle AXE = \angle C + \angle E$. Since Y lies between X and E, $\angle AXY = \angle AXE$.
- Triangle YDB: its angles at D and B are $\angle D$ and $\angle B$. A lies on the extension of DY beyond Y, so $\angle AYB = \angle D + \angle B$, and $\angle AYX = \angle AYB$.
- Triangle AXY has angles $\angle A$, $\angle AXY$, $\angle AYX$, so

$$\angle A + (\angle C + \angle E) + (\angle B + \angle D) = 180°. \quad \square$$

(Checked numerically with an irregular convex pentagon: the sum came out at 180°.)

4. (a) Yes: SAS (ZHZ). $\angle B$ lies between AB and BC.

(b) Yes: ASA (HZH). AB lies between angles A and B.

(c) Not necessarily. $\angle B$ is not between AB and AC; it lies opposite AC, and AC is the shorter given side. That is the ambiguous SSA case: there can be two different triangles (see the counterexample in notes §4).

(d) Yes: RHS (ZZR). Hypotenuse $AB = DE$ and leg $AC = DF$.

(e) No. Equal angles only give similar triangles; one can be an enlargement of the other.

(f) Yes: AAS (ZHH). BC is not between angles A and B, but the third angles are also equal, so this reduces to ASA.

5. (a)

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Gegeven:      triangle ABC with AB = AC; M is the midpoint of BC
Te bewijzen:  AM is perpendicular to BC, and angle BAM = angle CAM

Statement                                   Reason
1. AB = AC                                  given
2. BM = CM                                  M is the midpoint of BC
3. AM = AM                                  common side
4. triangle ABM congruent to triangle ACM   SSS (ZZZ), from 1, 2, 3
5. angle AMB = angle AMC                    corresponding angles of 4
6. angle AMB + angle AMC = 180              B, M, C on one line (straight angle)
7. angle AMB = angle AMC = 90               from 5 and 6
8. so AM is perpendicular to BC             definition of perpendicular
9. angle BAM = angle CAM                    corresponding angles of 4
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So AM is perpendicular to BC and bisects $\angle A$. □

(b) By Thales, $\angle ACB = 90°$, so $\angle ABC = 180° - 90° - 34° = 56°$. Triangle OAC is isosceles ($OA = OC$), so $\angle OCA = 34°$ and $\angle AOC = 180° - 68° = 112°$. Then $\angle BOC = 180° - 112° = 68°$ (straight angle along diameter AB). (Alternatively, triangle OBC is isosceles with base angles 56°, so $\angle BOC = 180° - 112° = 68°$.) $BC = \sqrt{10^2 - 6^2} = \sqrt{64} = 8$, using the right angle at C.

6. (a) $AB = \sqrt{7^2 + 24^2} = \sqrt{625} = 25$.

(b) $AC = \sqrt{13^2 - 5^2} = \sqrt{144} = 12$.
(c) $AB = \sqrt{25 + 64} = \sqrt{89} \approx 9.43$.

7. In each case let c be the longest side and compare $a^2 + b^2$ with c^2.

(a) $20^2 + 21^2 = 400 + 441 = 841 = 29^2$: right.
(b) $8^2 + 15^2 = 289 > 256 = 16^2$: acute.
(c) $6^2 + 7^2 = 85 < 100 = 10^2$: obtuse.
(d) $(\sqrt{5})^2 + (\sqrt{11})^2 = 5 + 11 = 16 = 4^2$: right.

8. (a) $\frac{9}{6} = \frac{12}{8} = \frac{15}{10} = 1.5$, so $\triangle ABC \sim \triangle DEF$ (zzz) with scale factor $k = 1.5$.

(b) $AB^2 + BC^2 = 36 + 64 = 100 = AC^2$, so by the converse of Pythagoras the angle at B is 90°. Area $= \frac{1}{2} \cdot 6 \cdot 8 = 24$.

(c) Way 1: area factor $k^2 = 2.25$, so area $= 2.25 \cdot 24 = 54$. Way 2: DEF is right-angled at E (the vertex matching B), so area $= \frac{1}{2} \cdot 9 \cdot 12 = 54$.

9. (a) $\angle A$ is common, and $\angle ADE = \angle ABC$ (F-angles, $DE \parallel BC$). So the triangles are similar (hh).

(b) $AB = 4 + 6 = 10$, so the scale factor from ADE to ABC is $\frac{10}{4} = 2.5$. Then $BC = 2.5 \cdot 5 = 12.5$, $AC = 2.5 \cdot 3 = 7.5$, and $EC = 7.5 - 3 = 4.5$.

10. (a) $\angle ASB = \angle DSC$ (vertically opposite) and $\angle SAB = \angle SDC$ (Z-angles, $AB \parallel CD$, transversal AD). So the triangles are similar (hh), with $A \leftrightarrow D$, $B \leftrightarrow C$.

(b) Scale factor $\frac{SD}{SA} = \frac{9}{6} = 1.5$. So $CD = 1.5 \cdot 8 = 12$ and $SC = 1.5 \cdot 5 = 7.5$.

11. BC is opposite A and AC is adjacent: $\tan 35° = \frac{BC}{12}$, so $BC = 12 \tan 35° \approx 8.40$. $\cos 35° = \frac{12}{AB}$, so $AB = \frac{12}{\cos 35°} \approx 14.65$.

12. Let the angle opposite the leg 5 be α. $\tan\alpha = \frac{5}{9}$, so $\alpha \approx 29.1°$, and the other angle is $90° - \alpha \approx 60.9°$. Hypotenuse $\sqrt{25+81} = \sqrt{106} \approx 10.30$.

13. (a) $\left(\frac{1}{2}\sqrt{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$.

(b) $\frac{1}{3}\sqrt{3} \cdot \sqrt{3} = \frac{3}{3} = 1$.

(c) $\frac{1}{2}\sqrt{3} \cdot \frac{1}{2}\sqrt{3} - \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$.

(d) $\cos^2\alpha = 1 - \frac{9}{25} = \frac{16}{25}$, and $\cos\alpha > 0$ for an acute angle, so $\cos\alpha = \frac{4}{5}$. Then $\tan\alpha = \frac{3/5}{4/5} = \frac{3}{4}$. (A 3-4-5 triangle.)

14. $k = 200$.

(a) Area factor $k^2 = 40\,000$: $50 \cdot 40\,000 = 2\,000\,000$ cm² $= 200$ m² (because 1 m² $= 10\,000$ cm²).

(b) Volume factor $k^3 = 8\,000\,000$: $1500 \cdot 8\,000\,000 = 1.2 \cdot 10^{10}$ cm³ $= 12\,000$ m³ (because 1 m³ $= 1\,000\,000$ cm³).

School-exam level

15. The wall, the ground and the ladder form a right triangle with hypotenuse 6.50.

(a) Height $= \sqrt{6.50^2 - 1.80^2} = \sqrt{39.01} \approx 6.25$ m.

(b) $\cos\alpha = \frac{1.80}{6.50}$, so $\alpha \approx 73.9°$.

(c) New height $= \sqrt{6.50^2 - 2.30^2} = \sqrt{36.96} \approx 6.08$ m. The top slides down $\sqrt{39.01} - \sqrt{36.96} \approx 0.17$ m.

(d) Before: $73.9°$ lies between 70° and 76°, so it is safe. After: $\cos\alpha = \frac{2.30}{6.50}$ gives $\alpha \approx 69.3° < 70°$: not safe.

16. (a) At most 1 m up per 12 m horizontally, so the horizontal length is at least $12 \cdot 0.45 = 5.40$ m.

(b) $\sqrt{5.40^2 + 0.45^2} \approx 5.42$ m.

(c) $\tan\alpha = \frac{0.45}{5.40} = \frac{1}{12}$, so $\alpha \approx 4.8°$. As a percentage: $\frac{1}{12} \cdot 100\% \approx 8.33\%$.

17. (a) Let F be the foot of the tower and T the top. In the right triangle AFT: $\tan 32° = \frac{h}{AF}$, so $AF = \frac{h}{\tan 32°}$. In BFT: $BF = \frac{h}{\tan 51°}$. Since $AF - BF = AB = 40$:

$$\frac{h}{\tan 32°} - \frac{h}{\tan 51°} = 40.$$

(b) Factor out h:

$$h = \frac{40}{\dfrac{1}{\tan 32°} - \dfrac{1}{\tan 51°}} \approx 50.60 \text{ m}.$$

$BF = \frac{h}{\tan 51°} \approx 40.97$ m. (Check: $AF \approx 80.97$ m, and $80.97 - 40.97 = 40$.)

18. (a) $AC = \sqrt{60^2 + 40^2} = \sqrt{5200} \approx 72.11$ cm.

(b) $CG \perp$ bottom face, so triangle ACG is right-angled at C: $AG = \sqrt{5200 + 30^2} = \sqrt{6100} \approx 78.10$ cm.

(c) The longest segment on the bottom is the diagonal, 72.11 cm < 75 cm, so the rod cannot lie flat. The longest segment in the box is the space diagonal, 78.10 cm > 75 cm, so the rod does fit, slanting along (near) the space diagonal.

(d) The angle is $\angle GAC$: $\tan\angle GAC = \frac{30}{\sqrt{5200}}$, so the angle is about 22.6°.

19. (a) $MN = 3$ (half the side) and $TM = 4$, with a right angle at M: $TN = \sqrt{9+16} = 5$. $AC = 6\sqrt{2}$, so $AM = 3\sqrt{2}$ and $TA = \sqrt{(3\sqrt{2})^2 + 4^2} = \sqrt{18+16} = \sqrt{34} \approx 5.83$.

(b) The projection of TA on the base is MA, so the angle is $\angle TAM$ with $\tan\angle TAM = \frac{4}{3\sqrt{2}}$: about 43.3°.

(c) $TN \perp BC$ (triangle TBC is isosceles, N the midpoint) and $MN \perp BC$, so the angle is $\angle TNM$ with $\tan\angle TNM = \frac{4}{3}$: about 53.1°.

(d) Base $6 \cdot 6 = 36$; four faces of $\frac{1}{2} \cdot 6 \cdot 5 = 15$ each. Total $36 + 60 = 96$.

(e) Front view: an isosceles triangle with base 6 (edge AB; CD is hidden behind it) and height 4; its slanted sides have length $\sqrt{3^2+4^2} = 5$. Top view: a square 6×6 with both diagonals drawn, crossing at the point T (above M); each half-diagonal has length $3\sqrt{2} \approx 4.24$. The slanted sides of the front view are the images of the edges TA and TB (with TD and TC hidden behind them). They have length 5, not the true edge length $TA = \sqrt{34} \approx 5.83$: the edges slope away from the viewer, are not parallel to the front plane, and so appear shorter. In the view the apex is 3 to the side of A and 4 above it, so the slanted side is $\sqrt{3^2+4^2} = 5$. This equals the true length of TN for the side faces TBC and TAD, because TN of those faces lies in a plane parallel to the front plane.

20. Side view: Lisa's eye L at height 4.50, the top of the wall W at height 3.00, at horizontal distance 6.00.

(a) The sight line over W drops $4.50 - 3.00 = 1.50$ m over 6.00 m, that is 0.25 m per metre. From W it must drop another 3.00 m to reach the ground: that takes $3.00/0.25 = 12.00$ m. The blind spot is 12.00 m deep. (Similar triangles: $\frac{d}{6} = \frac{3}{1.5}$.)

(b) At 5.00 m behind the wall, the sight line is at height $3.00 - 5 \cdot 0.25 = 1.75$ m. Tom is 1.80 m, so Lisa sees the top $1.80 - 1.75 = 0.05$ m (5 cm) of his head.

(c) The sight line must go through the top of the wall (height 3.00, at 6.00 m) and hit the ground 4.00 m behind the wall. So it drops 3.00 m over 4.00 m, i.e. 0.75 m per metre. Going back 6.00 m to Lisa it rises $6 \cdot 0.75 = 4.50$ m, so her eyes must be at $3.00 + 4.50 = 7.50$ m.

21. (a) Let s be the shadow length. The lamp, the foot of the lamppost and the tip of the shadow form a big right triangle; Ella and her shadow form a small one inside it (A-figure, both vertical). So

$$\frac{1.60}{4.80} = \frac{s}{s+3} \implies 3s = s+3 \implies s = 1.50 \text{ m}.$$

(b) With distance d: $\frac{1.60}{4.80} = \frac{s}{s+d}$ gives $3s = s + d$, so $s = \frac{1}{2}d$, for every d.

22. (a) Join the centre O to all six vertices. The six triangles are congruent (SSS: two radii and one side each), so the six angles at O are equal: $360°/6 = 60°$. Each triangle is isosceles with apex angle 60°, so its base angles are $\frac{180°-60°}{2} = 60°$: equilateral.

(b) Height $= \sqrt{4^2-2^2} = \sqrt{12} = 2\sqrt{3}$. Area of one triangle $= \frac{1}{2} \cdot 4 \cdot 2\sqrt{3} = 4\sqrt{3}$. Hexagon: $6 \cdot 4\sqrt{3} = 24\sqrt{3} \approx 41.57$.

(c) Two opposite sides are the bases of two opposite triangles, and the heights from O to those sides lie on one line (both are perpendicular to parallel sides through O). Distance $= 2 \cdot 2\sqrt{3} = 4\sqrt{3} \approx 6.93$.

23. (a) Triangle ABC has no right angle (its angles are 40°, 65° and 75°). SOS-CAS-TOA only applies in right triangles; use the right triangles ADC and BDC.

(b) In ADC: $\tan 40° = \frac{5}{AD}$, so $AD = \frac{5}{\tan 40°} \approx 5.96$. In BDC: $DB = \frac{5}{\tan 65°} \approx 2.33$. $AB = AD + DB \approx 8.29$ (computed from unrounded values: 8.2903...).

(c) $AC = \frac{5}{\sin 40°} \approx 7.78$; $BC = \frac{5}{\sin 65°} \approx 5.52$. Area $= \frac{1}{2} \cdot AB \cdot CD \approx \frac{1}{2} \cdot 8.2903 \cdot 5 \approx 20.73$.

Challenge

24. ($\Rightarrow$) Since $\angle C = 90°$, the converse of Thales' theorem says C lies on the circle with diameter AB. Its centre is the midpoint M and its radius is $\frac{1}{2}AB$. So $MC = \frac{1}{2}AB$.

A direct proof without the converse of Thales: reflect C in M to get D. As in notes §6, $ACBD$ is a rectangle, its diagonals are equal ($CD = AB$) and they bisect each other, so $MC = \frac{1}{2}CD = \frac{1}{2}AB$.

($\Leftarrow$) Suppose $MC = \frac{1}{2}AB$. Then $MA = MB = MC$. Triangle MAC is isosceles, so $\angle MAC = \angle MCA =: \alpha$; triangle MBC is isosceles, so $\angle MBC = \angle MCB =: \beta$. The angle sum in ABC gives $\alpha + \beta + (\alpha + \beta) = 180°$, so $\angle C = \alpha + \beta = 90°$. □

25. (a) Let the line through C parallel to AD meet line BA (extended beyond A) at E. Write $\angle BAD = \angle DAC = \varphi$.

1. $\angle AEC = \angle BAD = \varphi$ (F-angles: $AD \parallel EC$, transversal BE).
2. $\angle ACE = \angle DAC = \varphi$ (Z-angles: $AD \parallel EC$, transversal AC).
3. So triangle ACE has two equal angles at E and C; by the converse of the isosceles triangle theorem, $AE = AC$.
4. In triangle BEC, $AD \parallel EC$ with A on BE and D on BC: an A-figure. So $\triangle BAD \sim \triangle BEC$, giving $\frac{BA}{BE} = \frac{BD}{BC}$. Hence $\frac{BA}{BE-BA} = \frac{BD}{BC-BD}$, i.e. $\frac{BA}{AE} = \frac{BD}{DC}$. (Taking reciprocals: $\frac{BE}{BA} = \frac{BC}{BD}$, subtract 1 from both sides.)
5. With step 3: $\dfrac{BD}{DC} = \dfrac{AB}{AE} = \dfrac{AB}{AC}$. □

(b) $\frac{BD}{DC} = \frac{6}{9} = \frac{2}{3}$ and $BD + DC = 10$, so $BD = \frac{2}{5} \cdot 10 = 4$ and $DC = 6$. (Verified with coordinates in SymPy.)

26. Let O be the intersection of the perpendicular bisectors of AB and BC. (They intersect: if they were parallel, AB and BC would be parallel, but they meet at B and are not on one line.)

1. O is on the perpendicular bisector of AB, so $OA = OB$ (notes §5, $\Rightarrow$).
2. O is on the perpendicular bisector of BC, so $OB = OC$.
3. So $OA = OC$, and by the converse ($\Leftarrow$ in notes §5) O lies on the perpendicular bisector of AC.

So all three perpendicular bisectors pass through O. Since $OA = OB = OC$, the circle with centre O and radius OA passes through B and C: the circumscribed circle (*omgeschreven cirkel*). □

27. (a) Let the other leg be b and the hypotenuse c. Then $c^2 - b^2 = 144$, so $(c-b)(c+b) = 144$ with $0 < c - b < c + b$. Since $(c+b) - (c-b) = 2b$ is even, both factors have the same parity; their product is even, so both are even. The even factor pairs of 144 with the smaller factor first are $2 \cdot 72$, $4 \cdot 36$, $6 \cdot 24$, $8 \cdot 18$ ($12 \cdot 12$ would give $b = 0$). Solving $c - b = s$, $c + b = t$ gives $c = \frac{s+t}{2}$, $b = \frac{t-s}{2}$:

$c-b$	$c+b$	b	c	triangle
2	72	35	37	12, 35, 37
4	36	16	20	12, 16, 20
6	24	9	15	9, 12, 15
8	18	5	13	5, 12, 13

There are exactly four. (A computer search over all $b, c < 200$ found the same four.)

(b) $(m^2-n^2)^2+(2mn)^2 = m^4-2m^2n^2+n^4+4m^2n^2 = m^4+2m^2n^2+n^4 = (m^2+n^2)^2$, and all three numbers are positive whole numbers when $m > n \geq 1$. $\square$ $m=6, n=1$ gives $(35,12,37)$; $m=3, n=2$ gives $(5,12,13)$; $m=4, n=2$ gives $(12,16,20)$. The triple $(9,12,15)$ does not come from the formula directly (no m, n give a 9 or a 15 together with a 12); it is $3\cdot(3,4,5)$, with $(3,4,5)$ from $m=2, n=1$. So you need the formula *times a factor* to get every triple.

28. Take cube $ABCD.EFGH$ with E, F, G, H above A, B, C, D. The space diagonals AG and CE both lie in the plane through A, C, G, E. In that plane $ACGE$ is a rectangle: $AC = a\sqrt{2}$ (face diagonal) and $CG = AE = a$, with right angles because the vertical edges are perpendicular to the bottom face. Its diagonals AG and CE are equal and bisect each other at the centre S of the cube.

In the right triangle ACG: $\tan\angle GAC = \frac{a}{a\sqrt{2}} = \frac{1}{\sqrt{2}}$, so $\angle GAC \approx 35.26°$. Triangle SAC is isosceles ($SA = SC$), so $\angle SCA = \angle SAC \approx 35.26°$ and

$$\angle ASC = 180° - 2\cdot 35.26° \approx 109.5°.$$

The acute angle between the diagonals is $180° - 109.47° \approx 70.5°$. (These agree with the exact value $\cos^{-1}\frac{1}{3} \approx 70.53°$, checked in SymPy.) The answer does not depend on a because all cubes are similar: a cancels in $\frac{a}{a\sqrt{2}}$, and angles are unchanged by enlargement.

29. (a) Triangles ADC and CDB are both right-angled at D. $\angle ACD = 90° - \angle A$ (angle sum in ADC) and $\angle B = 90° - \angle A$ (angle sum in ABC), so $\angle ACD = \angle B$. Thus $\triangle ADC \sim \triangle CDB$ (hh), with $A \leftrightarrow C$, $D \leftrightarrow D$, $C \leftrightarrow B$. So

$$\frac{AD}{CD} = \frac{CD}{BD} \implies \frac{q}{h} = \frac{h}{p} \implies h^2 = pq. \quad \square$$

(b) Compute the area of ABC in two ways: with base BC and height AC (the legs are perpendicular), area $= \frac{1}{2}ab$; with base AB and height CD, area $= \frac{1}{2}ch$. So $ab = ch$. $\square$

(c) $h = \sqrt{9\cdot 4} = 6$ and $c = 13$. From notes §7, $a^2 = cp = 13\cdot 9 = 117$, so $a = \sqrt{117} = 3\sqrt{13}$; $b^2 = cq = 52$, so $b = 2\sqrt{13}$. (Or: $a = \sqrt{p^2+h^2} = \sqrt{81+36}$.) Check (b): $ab = 3\sqrt{13}\cdot 2\sqrt{13} = 6\cdot 13 = 78$ and $ch = 13\cdot 6 = 78$.

30. Let the incircle touch BC at P, CA at Q and AB at R.

(a) Take vertex A with tangent points Q and R. The radius to a point of tangency is perpendicular to the tangent, so $\angle AQI = \angle ARI = 90°$. Triangles AQI and ARI have a right angle, the common hypotenuse AI and $IQ = IR = r$. By RHS (ZZR) they are congruent, so $AQ = AR$. The same works at B and C. $\square$

(b) Quadrilateral $CPIQ$ has right angles at C (given), P and Q (radius perpendicular to tangent), so the fourth angle is $360° - 270° = 90°$ too: it is a rectangle, and opposite sides of a rectangle are equal, so $CP = IQ = r$ and $CQ = IP = r$. Now

$$a + b = (CP + PB) + (CQ + QA) = 2r + PB + QA = 2r + BR + AR = 2r + c,$$

using (a) for $PB = BR$ and $QA = AR$. So $r = \dfrac{a+b-c}{2}$. $\square$

(c) $\frac{3+4-5}{2} = 1$, $\frac{5+12-13}{2} = 2$, $\frac{8+15-17}{2} = 3$. (Checked against $r = \frac{\text{area}}{\text{half the perimeter}}$.) For Pythagorean triples the inradius is always a whole number: $a+b-c$ is always even (since $a^2+b^2=c^2$, $a+b$ and c have the same parity). For these three triangles it happens to be 1, 2, 3.